MIL-C-3098/67B(EL)
23 June 1975
SUPERSEDING
MIL-C-3098/67A(EL)
29 May 1969

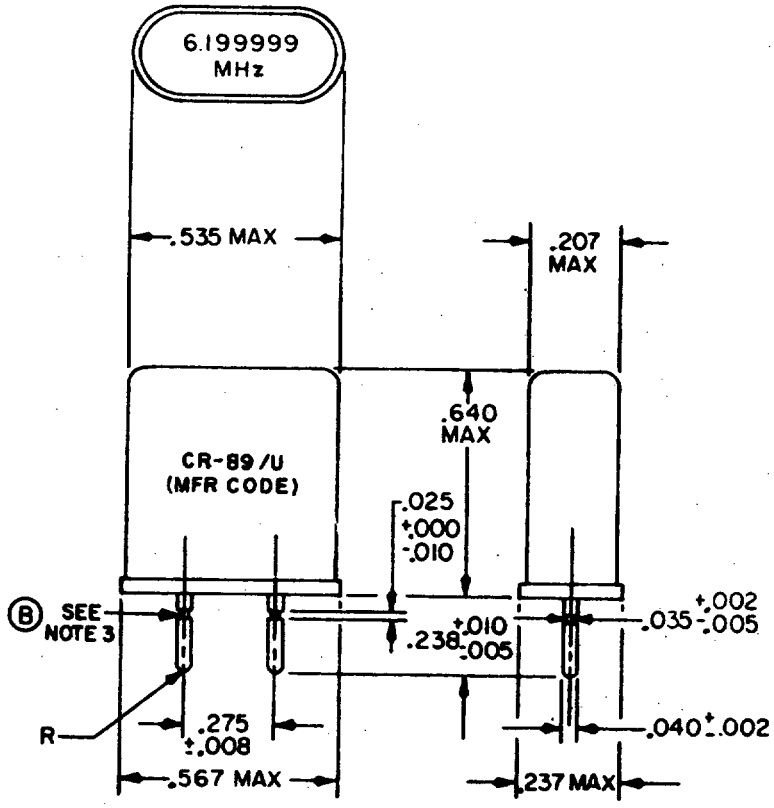
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-89/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 2.12 to 6.2 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.207	5.26
.005	.13	.237	6.02
.008	.20	.238	6.05
.010	.25	.275	6.99
.015	.38	.535	13.59
.025	.64	.567	14.40
.035	.89	.640	16.26
.040	1.02		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
Ⓑ 3. The pin undercut may be omitted.

Ⓑ denotes change

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-32/U (MIL-H-10056/18)
Dimensions, marking, and configuration	See figure
Frequency range	2.12 to 6.2 MHz, inclusive
Capacitance, shunt	7.0 pF, maximum
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See table II
Antiresonance, load capacitance	30.0 ±0.5 pF
Operating temperature range	-55° to +90° C, inclusive
Frequency tolerance	±50 PPM
Equivalent resistance	See table III
Shock: (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±15 percent
Vibration:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±15 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±15 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values		Rated drive level
		Resistance	Crystal current	
	MHz	Ohms	mA	mW
TS-330/TSM	2.12 to 2.25	50	15	10.0 ± 2.0
	2.25+ to 3.41	40	15	10.0 ± 2.0
	3.41+ to 6.20	25	20	10.0 ± 2.0

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TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
2.12 to 2.25	500
2.25+ to 2.80	320
2.80+ to 3.41	250
3.41+ to 4.10	180
4.10+ to 5.10	120
5.10+ to 6.20	75

Custodian:
Army - EL

Review activities:
Army - MI, MU
DSA - ES

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-A172-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-3098/67B(EL) CRYSTAL UNIT, QUARTZ, CR-89/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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